Cycle Test 1: worked solutions

26 Aug 2024 · 20 marks · 1 hour (open code book) · CEPC14

Q1

6 marks

> Fill in the blanks:
>
> (i) The mineral names (not the full form or chemical compound name) of C_2S and C_3S are ____ and ____ respectively.
>
> (ii) According to IS 456:2000, the maximum aggregate size should not be more than ____ of the member size, nor more than the space between reinforcement bars minus ____ mm.
>
> (iii) In the rotary kiln, the powdered constituents enter from the ____ end and exit from the ____ end.
>
> (iv) The needle-like crystals in the hydrated cement paste formed at the early hydration stage are called ____.
>
> (v) The water used for concreting should not have pH less than ____.
>
> (vi) Modern Portland cement was invented by ____.

1. **Belite** and **alite**.
2. **One-fourth (1/4)** of the minimum thickness of the member; clear spacing between bars minus **5 mm** (IS 456 Cl. 5.3.3).
3. Enter at the **upper (feed, cooler)** end; exit at the **lower (burner, hotter)** end. The kiln is inclined and the flame is at the lower end.
4. **Ettringite** (AF_t).
5. **6** (IS 456 Cl. 5.4.2: pH not less than 6).
6. **Joseph Aspdin** (patent of 1824). (His son William Aspdin and later I. C. Johnson, who first burnt the mix to clinkering temperature in 1845, developed it towards today's cement.)

Q2

2 marks

> Write the chemical reaction equations for (i) formation of C_3S from limestone and silica, (ii) hydration of C_3S.

(i) Formation in the kiln

$$CaCO_3 \rightarrow CaO + CO_2 \text{ (decarbonation, 700–900 °C)}$$

$$2CaO + SiO_2 \rightarrow 2CaO{\cdot}SiO_2 \ (C_2S,\ 900–1200\ °C)$$

$$2CaO{\cdot}SiO_2 + CaO \rightarrow 3CaO{\cdot}SiO_2 \ (C_3S, \text{ above about } 1250–1300\ °C \text{ with melt})$$

Overall: **$3CaCO_3 + SiO_2 \rightarrow 3CaO{\cdot}SiO_2 + 3CO_2$**

(ii) Hydration

$$2(3CaO{\cdot}SiO_2) + 6H_2O \rightarrow 3CaO{\cdot}2SiO_2{\cdot}3H_2O + 3Ca(OH)_2$$

In short notation: **$2C_3S + 6H \rightarrow C_3S_2H_3$ (C-S-H) + 3CH**.

Q3

4 marks

> Compare: (i) constituents, setting times and strength of Ordinary Portland Cement and Low Heat Cement, (ii) mechanism of action of air entraining admixtures and water reducing admixtures.

(i) OPC vs low heat Portland cement

	OPC (IS 269)	Low heat cement (IS 12600)
Constituents	C_3S 45–60%, C_2S 15–30%, C_3A 6–12%	**Less C_3S (25–35%) and C_3A, more C_2S (40–50%)**, more C_4AF
Fineness (min.)	225 m^2/kg	320 m^2/kg
Initial setting time	≥ 30 min	**≥ 60 min**
Final setting time	≤ 600 min	≤ 600 min
Compressive strength (MPa)	OPC 43: 23 (3 d), 33 (7 d), 43 (28 d)	**10 (3 d), 16 (7 d), 35 (28 d)**: slower gain, similar or higher ultimate strength

	OPC (IS 269)	Low heat cement (IS 12600)
Heat of hydration	High	**≤ 272 kJ/kg at 7 d, ≤ 314 kJ/kg at 28 d**
Use	General construction	Mass concrete: dams, thick rafts

(ii) Mechanism

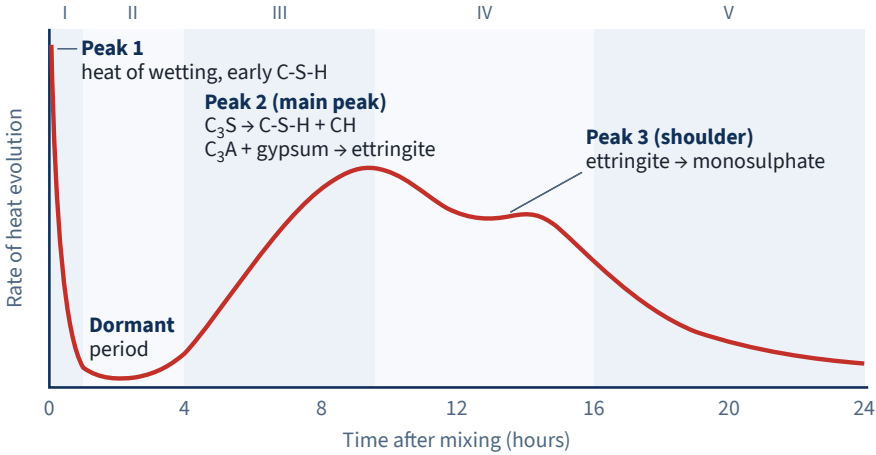
	Air-entraining admixture (AEA)	Water-reducing admixture
Chemicals	Surfactants: vinsol resin, salts of wood resins, fatty acids, sulfonated hydrocarbons	Lignosulphonates, hydroxycarboxylic acids (normal); SNF, SMF, PCE (superplasticizers)
Where it acts	**Air–water interface**	**Cement–water interface**
Mechanism	Molecules have a hydrophilic head and hydrophobic tail; they gather at air–water interfaces, **lower surface tension**, and **stabilize tiny bubbles (about 10–100 µm)** formed during mixing so they don't merge or escape. Charged films on bubbles and cement grains keep them dispersed	Molecules **adsorb on cement grains**, give them a like (negative) charge → **electrostatic repulsion** (and **steric hindrance** for PCE); flocs break up and **trapped water is released**; lower surface tension also helps wetting
Result	4–6% air, spacing factor < 0.2 mm → **freeze–thaw resistance**, better workability and cohesion, less bleeding; some loss of strength (about 5% per 1% air)	**Same workability with less water** (5–40%) → higher strength and durability, or more workability at same w/c

Q4

4 marks

> Discuss (i) the evolution of heat of hydration of cement with time, with the related figure, (ii) determination of the amount of deleterious substances in aggregates by the sedimentation method.

(i) Heat of hydration



- **Stage I (minutes)**: sharp first peak: heat of wetting, rapid dissolution, early ettringite.
- **Stage II, dormant period (1–3 h)**: very little heat; a layer on the grains slows reaction; concrete stays workable.
- **Stage III, acceleration (3–10 h)**: main peak from **C_3S hydration** (C-S-H + CH); initial and final set occur here.
- **Stage IV, deceleration**: a shoulder at 12–20 h as **ettringite converts to monosulphate** when gypsum runs out.
- **Stage V**: slow, diffusion-controlled reaction for months.
- Bogue: about 50% of the total heat in 1–3 days, 75% in 7 days, 83–91% in 6 months. Compounds: C_3A 867, C_3S 502, C_4AF 419, C_2S 260 J/g.

(ii) Clay, fine silt and fine dust by the sedimentation method (IS 2386 Part 2)

Principle: particles settle in water at speeds given by **Stokes' law**, $v = (\rho_s - \rho_w)\, g\, d^2 / 18\eta$, so after a fixed time, liquid sampled at a fixed depth contains only particles finer than a chosen size (about **20 µm**).

Procedure:

1. Take about **300 g** of sand (or a weighed sample of coarse aggregate), W_1, in a glass jar with about 300 ml of **sodium oxalate solution (8 g/l)** as dispersing

agent.

2. Rotate/shake the jar for about 15 min so that the clay and silt separate from the grains.
3. Wash the fines into a 1000 ml measuring cylinder; make up the volume, invert several times, and let it stand.
4. After the time given by Stokes' law, draw **25 ml (V)** with a pipette from a depth of **10 cm**.
5. Evaporate and dry the sample at 100–110 °C; weigh the residue, W_2.

$$\text{Clay and fine silt (\%)} = \frac{100}{W_1} \times \left(\frac{1000\, W_2}{V} - 0.8 \right)$$

(0.8 g is the sodium oxalate in one litre.) IS 383 limits clay, fine silt and fine dust to about **3% in uncrushed sand** and **15% in crushed sand** by mass.

A simpler **field settling test**: shake sand in 1% salt solution in a measuring cylinder, allow 3 h, and measure the silt layer; it should not exceed about **8% by volume**.

Q5

4 marks

> In a Blaine air permeability test, 2.95 g of a cement of particle density 3.12 g/cc is taken in a bed in the cell of volume 1.9 cm^3. When the test is done on the reference cement of fineness 225 m^2/kg and particle density 3.15 g/cc, 28 seconds was needed for the liquid in the manometer to fall from the middle to the third level mark. If the time taken for the same liquid fall is 27 seconds for the sample cement, determine the fineness of the sample cement. Take viscosity of air at 29 °C as 1.84×10^{-5} Pa·s.
>
> Formulas: $m = \rho V(1 - e)$; $K = S_0\, \rho_0\, (1 - e)\, \sqrt{(0.1\, \eta_0)} / (\sqrt{e^3}\, \sqrt{t_0})$; $S = 521.08\, K\, \sqrt{t} / \rho$ (cm^2/g).

Porosity of the sample bed

$e = 1 - m / (\rho V) = 1 - 2.95 / (3.12 \times 1.9) = 1 - 0.4976 =$ **$0.502 \approx 0.50$**

So the bed has the standard porosity of 0.500, for which $S = 521.08\ K\sqrt{t}/\rho$ applies.

Apparatus constant (reference cement: $S_0 = 225\ m^2/kg = 2250\ cm^2/g$, $\rho_0 = 3.15$, $t_0 = 28$ s, $e = 0.5$)

$K = 2250 \times 3.15 \times (1 - 0.5) \times \sqrt{(0.1 \times 1.84 \times 10^{-5})} / (\sqrt{0.5^3} \times \sqrt{28})$

$= 3543.75 \times 1.3565 \times 10^{-3} / (0.35355 \times 5.2915)$

$= 4.8070 / 1.8708 =$ **2.569**

Fineness of the sample ($t = 27$ s, $\rho = 3.12$)

$S = 521.08 \times 2.569 \times \sqrt{27} / 3.12 = 521.08 \times 2.569 \times 5.196 / 3.12$

= **2230 cm²/g = 223 m²/kg**

Check: at equal porosity and viscosity, $S = S_0\ (\rho_0/\rho)\ \sqrt{(t/t_0)} = 225 \times (3.15/3.12) \times \sqrt{(27/28)} =$ **223 m²/kg** ✓.

The sample is **just below the IS 269 minimum of 225 m²/kg** for OPC; within the accuracy of the test it is borderline, so it should be re-tested.